

Sequential Algorithms for Sublinear Distance to Multicalibration

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1 Introduction

While a standard definition for perfect calibration is accepted, there is no such universal metric for miscalibration. Expected Calibration Error (ECE) is one common metric which measures the sum of biases across level sets, but is sensitive to small perturbations in predictions and is unobtainable at $O(\sqrt{T})$ rates, as noted in [7].

Calibration distance, which is the ℓ_1 distance between a given transcript and the nearest perfectly calibrated transcript (intuitively understood as the minimal “tweaking” necessary to calibrate a set of predictions p) overcomes these limitations: it is continuous with respect to the sequence of predictions p and in [1], Arunachaleswaran et al. give an efficient algorithm with distance to calibration at most $2\sqrt{T} + 1$.

Controlling calibration distance gives good marginal performance, but makes no conditional promises. In particular, there is no assurance that a predictor with small calibration distance is unbiased on arbitrary subsequences of the transcript, the problem this work addresses. Prior work ([5]) gives sublinear sequential “multi-miscalibration,” but as measured by the maximum across group-level-set intersections and is not easily transferrable to the calibration distance setting.

We bridge this gap by defining multicalibration distance (which we denote by MCDist) as the maximum ℓ_1 distance across groups to a predictor calibrated on that group. Then, we modify the AOSA algorithm of [1] to control MCDist at $O(\sqrt{|\mathcal{G}|T})$ rates, although we believe game-theoretic approaches might allow future works to obtain $O(\sqrt{\log(|\mathcal{G}|)T})$ MCDist .

2 Notation & Definitions

In this paper, we study the sequential binary prediction setting. At each time step $t \in [T]$, our forecaster observes context $x^t \in \mathcal{X}$ for some context space $\mathcal{X}$, then makes a prediction $p^t \in [0, \frac{1}{m}, \dots, 1]$. Upon observing this prediction, an adversary generates a binary outcome $y^t \in \{0, 1\}$. While p^t is typically permitted to be a distribution over the action space, note all algorithms in this paper are deterministic using the power of two choices, so we will limit the forecaster to only pure strategies.

We will let $\mathcal{G}$ denote our set of groups, $\mathcal{G}(x^t)$ to be the set of groups in effect on iteration t , and each $g \in \mathcal{G}$ be a membership function $g : \mathcal{X} \rightarrow \{0, 1\}$.

Definition 1 (Expected Calibration Error (ECE)).

$$\text{ECE}(p^{1:T}, y^{1:T}) = \sum_{p \in [0,1]} \left| \sum_{t=1}^T \mathbb{1}[p^t = p](p^t - y^t) \right|$$

Note that we will often suppress the notation $p^{1:T}, y^{1:T}$ by instead writing p, y .

Definition 2 (Calibration). *A sequence of predictions $p^{1:T}$ is perfectly calibrated to labels $y^{1:T}$ iff. it satisfies $\text{ECE}(p, y) = 0$.*

Definition 3 (Multicalibration). *Fix some set of groups $\mathcal{G}$ (each day, the adversary can choose an arbitrary subset of these). A sequence of predictions $p^{1:T}$ is perfectly multicalibrated iff. the following holds:*

$$\forall g \in \mathcal{G}, \sum_{p \in [0,1]} \left| \sum_{t=1}^T g(x^t) \mathbb{1}[p^t = p](p^t - y^t) \right| = 0.$$

Note that the definition above can be extended by considering $g : \mathcal{X} \rightarrow [0, 1]$, i.e., examples with “partial” group membership.

The bias conditional on a prediction and group $V_T^{g,i} := V_T^{g,i}(\pi^s)$ is defined as

$$V_T^{g,i} = \sum_{t=1}^T \mathbb{1}[g(x^t) = 1, p^t = \frac{i}{m}](y^t - p^t)$$

Definition 4 (Calibration Distance (CalDist)). *Following [2] and [7], we define the calibration distance of a sequence of predictions $\mathcal{E}$ outcomes is the minimum ℓ_1 distance to a perfectly calibrated sequence, i.e.*

$$\text{CalDist}(p, y) = \min_{q \in \mathcal{C}} \|p - q\|_1$$

where $\mathcal{C} = \mathcal{C}(y)$ is the set of all perfectly calibrated sequences.

We define multicalibration distance of a sequence of predictions & outcomes as the natural extension of calibration distance, but for a collection of groups.

Definition 5 (Multicalibration Distance (MCDist)). *Specifically, the multicalibration distance is the maximum distance to calibration for any given group, or*

$$\text{MCDist}(p, y) = \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|p - q\|_1$$

where $\mathcal{C}_g = \mathcal{C}_g(y)$ is the set of all sequences of predictions that are perfectly calibrated on the iterations containing group g .

3 Main Results

3.1 Initial Lemmas

Lemma 3.1 ([7]). *For any sequences of predictions $p^{1:T}$ and outcomes $y^{1:T}$, $\text{CalDist}(p, y) \leq \text{ECE}(p, y)$.*

Proof. Define $q^{1:T}$ with $q^t = \bar{y}^T(p^t)$, where $\bar{y}^T(p) = \sum_{t=1}^T \frac{\mathbb{1}[p^t=p]}{\sum_{t=1}^T \mathbb{1}[p^t=p]} y^t$, i.e., the “average” label on days when p was played. Then,

$$\begin{aligned}
\text{CalDist}(p, y) &\leq \|p - q\|_1 \\
&= \sum_{t=1}^T |p^t - q^t| \\
&= \sum_{p \in [0,1]} \sum_{t=1}^T \mathbb{1}[p^t = p] |p - \bar{y}^T(p)| \\
&= \sum_{p \in [0,1]} |p - \bar{y}^T(p)| \sum_{t=1}^T \mathbb{1}[p^t = p] \\
&= \sum_{p \in [0,1]} \left| p \sum_{t=1}^T \mathbb{1}[p^t = p] - \bar{y}^T(p) \sum_{t=1}^T \mathbb{1}[p^t = p] \right| \\
&= \sum_{p \in [0,1]} \left| \sum_{t=1}^T \mathbb{1}[p^t = p] (p^t - y^t) \right| \\
&= \text{ECE}(p^{1:T}, y^{1:T}).
\end{aligned}$$

□

The above from [7].

Claim 1. *For any sequence of predictions $p = p^{1:T}$ and outcomes $y = y^{1:T}$, $\text{MCDist}(p, y) \leq \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}|$.*

Proof. By definition, we have that

$$\begin{aligned}
\text{MCDist}(p, y) &= \max_{g \in \mathcal{G}} \text{CalDist}(p, g, y) \\
&= \max_{g \in \mathcal{G}} \text{CalDist}(p^{S_g}, y^{S_g}) \\
&\leq \max_{g \in \mathcal{G}} \text{ECE}(p^{S_g}, y^{S_g}) && \text{by Lemma 3.1} \\
&= \max_{g \in \mathcal{G}} \sum_{i \in [m]} \left| \sum_{t=1}^T \mathbb{1}[g(x^t) = 1, p^t \in B(i)] (y^t - p^t) \right| \\
&= \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}|
\end{aligned}$$

□

We note the final line can be rewritten as $\max_{g \in \mathcal{G}} \text{ECE}(p, g)$, i.e. the term being maximized over is the ECE conditioned on indices with group g .

3.2 Warm up: disjoint groups

Theorem 3.2 (Informal). *In the case that the groups $\mathcal{G}$ partition the transcript π^T , running AOSA on each group simultaneously gives*

$$\text{MCDist}(p) = O(\sqrt{T})$$

Lemma 3.3. *Given some set of possibly overlapping groups $\mathcal{G}$, by taking*

$$\mathcal{G}' = \left\{ \bigcap_{G_i \in P} G_i \mid P \in \mathcal{P}(\mathcal{G}) \right\},$$

i.e., all possible intersections of groups in $\mathcal{G}$, Theorem 3.2 gives

$$\text{MCDist}(p) \leq |\mathcal{G}'| \sqrt{T} = O(\exp(|\mathcal{G}|) \sqrt{T})$$

Proof of Theorem 3.2. By assumption, the sets p^{S_g} partition p (and likewise for y^{S_g} , y). Thus, we may run AOSA separately on each $\pi^g := \{(p^t, y^t) \mid g(x^t) = 1\}$.

In this case, for each $g \in \mathcal{G}$, we can compute directly

$$\text{CalDist}(p^{S_g}, y^{S_g}) \leq 2\sqrt{|p^{S_g}|} + 1 \leq 2\sqrt{T} + 1.$$

□

Proof of Lemma 3.3. Fix some arbitrary set of (not necessarily disjoint) groups $\mathcal{G}$. For an arbitrary group $g \in \mathcal{G}$, note that

$$\begin{aligned} \text{CalDist}(p, g) &= \sum_{g' \in \mathcal{G}' \mid g \cap g' \neq \emptyset} \text{CalDist}(p, g') \\ &\leq \sum_{g' \in \mathcal{G}' \mid g \cap g' \neq \emptyset} (2\sqrt{T} + 1) \\ &\leq |\mathcal{G}'| (2\sqrt{T} + 1) \\ &= O(\exp(|\mathcal{G}|) \sqrt{T}). \end{aligned}$$

Since this holds for an arbitrary g , it holds for the group for which this quantity is maximum, giving us the desired result. □

3.3 Exponential Weights: $O(T^{3/4} \ln(|\mathcal{G}|T))$ MCDist

At a high level, we can obtain MCDist on the order of $T^{3/4} \ln(|\mathcal{G}|T)$ by using the sequential multicalibration algorithm in [5] to generate our look-ahead predictions in the AlmostOn-StepAhead algorithm [1].

Definition 6 (Surrogate Loss). For a fixed transcript π_s and parameter $\eta \in [0, 1)$, let surrogate multicalibration loss at iteration s be

$$L_s := L_s(\pi_s) = \sum_{g \in \mathcal{G}} \sum_{i \in [m]} e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}}$$

Observation 1. For any transcript π_s and $\eta \in [0, 1)$,

$$\max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}| \leq \frac{m}{\eta} \ln L_s$$

Proof. We proceed as follows:

$$\begin{aligned} \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}| &\leq m \max_{g \in \mathcal{G}} \max_{i \in [m]} |V_s^{g,i}| \\ &= \frac{m}{\eta} \ln \left(\exp(\eta \max_{g \in \mathcal{G}} \max_{i \in [m]} |V_s^{g,i}|) \right) \\ &= \frac{m}{\eta} \ln \left(\max_{g \in \mathcal{G}} \max_{i \in [m]} e^{\eta |V_s^{g,i}|} \right) \\ &\leq \frac{m}{\eta} \ln \left(\max_{g \in \mathcal{G}} \max_{i \in [m]} e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}} \right) \\ &\leq \frac{m}{\eta} \ln \left(\sum_{g \in \mathcal{G}} \sum_{i \in [m]} e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}} \right) \\ &= \frac{m}{\eta} \ln L_s \end{aligned}$$

□

We claim without proof that the bound in Observation 1 is tight up to an additive factor of $\frac{m}{\eta} \ln(m|\mathcal{G}|)$. This upper bound isn't needed in our proof, but can be found in [5].

Since MCDist is bounded above by the left term of Observation 1 and $\ln(x)$ is increasing in its argument, our goal is to show L_s can't grow too large throughout the course of the algorithm. We will do so by bounding the change in L_s at each iteration.

Lemma 3.4. Let $\Delta_s = \Delta_s(\pi^s) = L_s - L_{s-1}$ be the change in the surrogate loss at iteration s . Then

$$\Delta_s \leq \eta(y^s - p^s)C_s^i + \eta^2 L_s$$

where $C_s^i = \sum_{g \in \mathcal{G}(x^s)} \exp(\eta V_{s-1}^{g,i}) - \exp(-\eta V_{s-1}^{g,i})$.

Proof. At each iteration s , fix the prediction p^s . Observe the only change in L_s is due to the change in bias of groups $g \in \mathcal{G}(x^s)$ on prediction $i \in [m]$ corresponding to p^s . Thus, we have that

$$\Delta_s = L_s - L_{s-1}$$

$$\begin{aligned}
&= \sum_{g \in \mathcal{G}(x^s)} \left(e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}} \right) - \left(e^{\eta V_{s-1}^{g,i}} + e^{-\eta V_{s-1}^{g,i}} \right) \\
&= \sum_{g \in \mathcal{G}(x^s)} e^{\eta V_{s-1}^{g,i}} (e^{\eta(y^s - p^s)} - 1) + e^{-\eta V_{s-1}^{g,i}} (e^{-\eta(y^s - p^s)} - 1) \\
&\leq \sum_{g \in \mathcal{G}(x^s)} e^{\eta V_{s-1}^{g,i}} (\eta(y^s - p^s) + \eta^2(y^s - p^s)^2) + e^{-\eta V_{s-1}^{g,i}} (-\eta(y^s - p^s) + \eta^2(y^s - p^s)^2) \\
&\leq \eta^2 \sum_{g \in \mathcal{G}(x^s)} (e^{\eta V_{s-1}^{g,i}} + e^{-\eta V_{s-1}^{g,i}}) + \eta(y^s - p^s) \sum_{g \in \mathcal{G}(x^s)} (e^{\eta V_{s-1}^{g,i}} - e^{-\eta V_{s-1}^{g,i}}) \\
&= \eta^2 L_s + \eta(y^s - p^s) C_s^i
\end{aligned}$$

where the first inequality uses that $e^x \leq 1 + x + x^2$ for all $x \in [-1, 1]$, and the second uses that $|y^s - p^s| \leq 1 \implies (y^s - p^s)^2 \leq 1$. $\square$

We now leverage the power of two choices to eliminate the second term. To summarize, the algorithm AlmostOneStepAhead-Group picks two adjacent prediction candidates $i^*, i^* + 1$, arbitrarily predicts the first of the two, then picks a lookahead prediction from these two to guarantee $(y^s - \tilde{p}^s) C_s^i \leq 0$ for all $s \in [T]$.

Algorithm 1 AlmostOneStepAhead-Group

for $t = 1$ to T **do**

Observe x^t and for each $i \in [m]$, compute lookahead bias

$$V_{t-1}^{g,i} := V_{t-1}^{g,i}(\tilde{p}^{1:t-1}) = \sum_{s=1}^{t-1} \mathbb{1}(p^s = \frac{i}{m}, g(x^s) = 1)(y^s - \tilde{p}^s)$$

For all $i \in [m]$, compute

$$C_{t-1}^i = \sum_{g \in \mathcal{G}(x^t)} e^{\eta V_{t-1}^{g,i}} - e^{-\eta V_{t-1}^{g,i}}$$

Obtain any $i^* \in [m-1]$ such that $C_{t-1}^{i^*} \cdot C_{t-1}^{i^*+1} \leq 0$. Set $p^t = \frac{i^*}{m}$.

Set lookahead prediction $\tilde{p}^t$ to be $i' \in [i^*, i^* + 1]$ such that

$$(y^t - \tilde{p}^t) C_{t-1}^{i'} \leq 0$$

Observation 2. At any iteration $t \in [T]$, we note there will always exist an i^* such that $C_{t-1}^{i^*} \cdot C_{t-1}^{i^*+1} \leq 0$, as $V_s^{g,0} \geq 0$ and $V_s^{g,m} \leq 0$. Furthermore, there will always exist $i' \in [i^*, i^* + 1]$ such that $(y^t - \tilde{p}^t) C_{t-1}^{i'} \leq 0$: if $y^t = 1$, we set i' such that $C_{t-1}^{i'} \leq 0$, and if $y^t = 0$ we may obtain i' such that $C_{t-1}^{i'} \geq 0$.

We are now ready to prove our final bound:

Theorem 3.5 (AlmostOneStepAhead-Group guarantee). *With $\eta = \sqrt{\frac{\ln(2|\mathcal{G}|m)}{T}}$ and $m = (2T \ln(2|\mathcal{G}|T))^{1/4}$, AlmostOneStepAhead-Group yields a sequence of predictions $p = p^{1:T}$ fulfilling*

$$\text{MCDist}(p) = O((T \ln(|\mathcal{G}|m))^{3/4})$$

Proof. We first bound the surrogate loss L_T of our look ahead predictions $\tilde{p}$. Applying Lemma 3.4 on our look-ahead transcript,

$$\begin{aligned} L_T &= L_{T-1} + \Delta_{T-1} \\ &\leq L_{T-1} + \eta^2 L_{T-1} + \eta(y^T - \tilde{p}^T)C_T^i \\ &\leq L_{T-1}(1 + \eta^2) & (\eta(y^s - \tilde{p}^s)C_s^i \leq 0 \ \forall s) \\ &\leq e^{\eta^2} L_{T-1} \leq \dots \leq e^{T\eta^2} L_0 \end{aligned}$$

Now, $L_0 = \sum_{g \in \mathcal{G}} \sum_{i \in [m]} (e^0 + e^0) = 2|\mathcal{G}|m$, so we conclude $L_T \leq 2e^{T\eta^2} |\mathcal{G}|m$. Invoking Claim 1 and Observation 1 on our look ahead transcript, we obtain

$$\begin{aligned} \text{MCDist}(\tilde{p}, y) &\leq \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}| \leq \frac{m}{\eta} \ln L_T \\ &\leq m \left(T\eta + \frac{\ln(2|\mathcal{G}|m)}{\eta} \right) \\ &\leq m (2T \ln(2|\mathcal{G}|m))^{1/2} \end{aligned}$$

where the last inequality arises by setting $\eta = \sqrt{\frac{\ln(2|\mathcal{G}|m)}{T}}$.

Finally, note that the ℓ_1 distance between our look ahead predictions and our actual predictions at most $\frac{1}{m}$ at each iteration for a cumulative $\|p - \tilde{p}\|_1 \leq \frac{T}{m}$. Thus,

$$\begin{aligned} \text{MCDist}(p, y) &= \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|p - q\|_1 \\ &= \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|(p - \tilde{p}) + (\tilde{p} - q)\|_1 \\ &\leq \|p - \tilde{p}\|_1 + \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|\tilde{p} - q\|_1 & (\text{triangle inequality}) \\ &\leq \frac{T}{m} + \text{MCDist}(\tilde{p}, y) \\ &\leq \frac{T}{m} + m(2T \ln(2|\mathcal{G}|m))^{1/2} \\ &\leq 2(2T \ln(2|\mathcal{G}|T))^{3/4} \end{aligned}$$

where the last inequality arises by setting $m = (2T \ln(2|\mathcal{G}|T))^{1/4}$. □

It is likely the failure of the approach above to provide MCDist scaling with $\sqrt{T}$ is because we implemented a surrogate function tight to (α, m) -multicalibration, which fails to capture MCDist for the same discontinuity reason as ECE.

Since MCDist involves taking a maximum over groups, we certainly need a surrogate function to frame sequential predictions as a game. However, computing CalDist and MCDist

involves comparing the realized transcript to a hypothetical “closest” calibrated one, so it is not immediately clear how to formulate this as a convex-concave function.

This may require an approach similar to that of [7] in which we could bound MCDist by some group versions of lower calibration distance and smooth calibration error, although this is not an avenue we’ve thoroughly explored.

3.4 Greedy POTC: $O(\sqrt{T|\mathcal{G}|})$ MCDist

Rather than consider a holistic approach like AlmostOneStepAhead-Group, we can mimic the power of two choices algorithm from [4] by directly minimizing the bias contributed by each level set.

Observation 3. *We can extend Claim 1 as follows:*

$$\text{MCDist}(p, y) \leq \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_T^{g,i}| \leq \sum_{i \in [m]} \max_{g \in \mathcal{G}} |V_T^{g,i}|$$

Rather than bounding the group with the worst ECE, we can focus on undercutting the group with the worst bias within each level set. This gives rise to a new “greedy” algorithm that does just that:

Algorithm 2 GreedyGroupPOTC

for $t = 1$ to T **do**

Observe x^t , and define look ahead bias $V_{t-1}^{g,i} := V_{t-1}^{g,i}(\tilde{p}^{1:t-1})$ as

$$V_{t-1}^{g,i} = \sum_{s=1}^{t-1} \mathbb{1}(\tilde{p}^s = \frac{i}{m}, g(x^s) = 1)(y^s - \tilde{p}^s)$$

Pick index $i^* \in [m-1]$ fulfilling

$$V_{t-1}^{g(t,i^*),i^*} \cdot V_{t-1}^{g(t,i^*+1),i^*+1} \leq 0$$

for any $g(t,i) \in \arg \max_{g \in \mathcal{G}(x^t)} |V_t^{g,i}|$ that is a group with max absolute bias in level set i .

Predict $p^t = \frac{i^*}{m}$. Then, observe y^t , and set look ahead prediction $\tilde{p}^t = \frac{i'}{m}$ to fulfill

$$(y^t - \tilde{p}^t) V_{t-1}^{g(t,i'),i'} \leq 0$$

where $i' \in \{i^*, i^* + 1\}$

Observation 4. *There will always exist $i^* \in [m-1]$ such that the group with worst absolute bias in level set i^* has positive bias while $i^* + 1$ has negative bias, as $\forall g \in \mathcal{G}, V_t^{g,0} \geq 0$ and $V_t^{g,m} \leq 0$. Given this, if $y^t = 1$, note $y^t - \tilde{p}^t \geq 0$ for any choice of $\tilde{p}^t$, and there exists $i' \in \{i^*, i^* + 1\}$ such that $V_t^{g(t,i'),i'} \leq 0$; similarly, if $y^t = 0$, there exists $i' \in \{i^*, i^* + 1\}$ such that $V_t^{g(t,i'),i'} \geq 0$. In either case, we may pick look ahead prediction $\tilde{p}^t$ to fulfill $(y^t - \tilde{p}^t) V_{t-1}^{g(t,i'),i'} \leq 0$.*

At each iteration, we pick two adjacent level sets such that the groups in these level sets with the greatest absolute biases are of opposing signs. We arbitrarily predict one, then select our look ahead prediction to be the prediction in hindsight for which

$$|V_t^{g(t,i'),i'}| = |V_{t-1}^{g(t,i'),i'}| + (y^t - \tilde{p}^t) \leq |V_{t-1}^{g(t,i'),i'}|$$

While this decreases the bias of group $g(t,i')$, this may still *increase* the worst (absolute) bias across all groups within the level set. For example, if the algorithm picks a level set i for which there are two groups with max absolute bias, one with positive bias and one negative, the max absolute bias will always increase if both groups are included in $\mathcal{G}(x^t)$. Below is a hypothetical illustration:

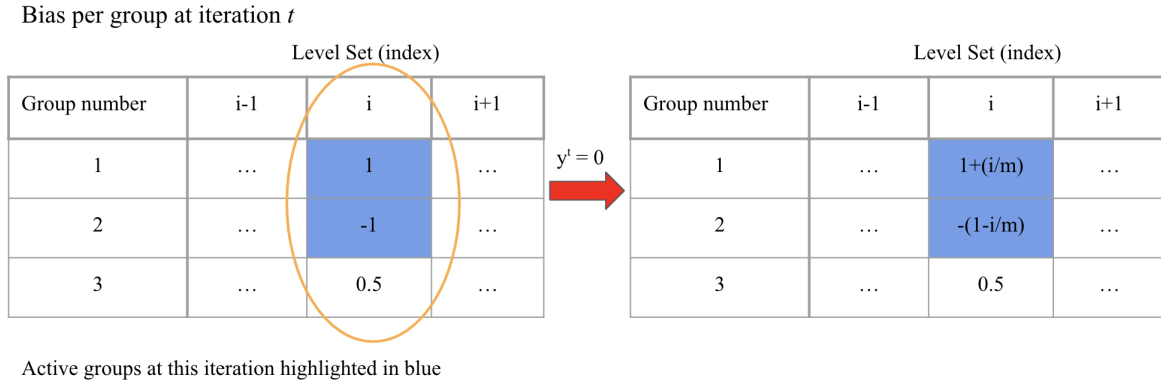


Figure 3.4: for some $t \in [T]$, each table contains the values of $V_t^{g,i}$. At this iteration, only groups 1, 2 are active. In level set i , both 1 and 2 are groups of max absolute bias at this iteration. Should GreedyGroupPOTC select level set i , the max bias will increase no matter whether the adversary picks label 0 or 1;

Even so, we claim any level set obtains max bias across its group at most $|\mathcal{G}|$. This is ultimately tight: we will state without further proof that an adversary knowledgeable of the GreedyGroupPOTC algorithm can induce a level set to reach $|\mathcal{G}|$ bias.

Lemma 3.6 (Worst group bias in a level set). *For any $g \in \mathcal{G}$ and $i \in [m]$, at any iteration $t \in [T]$ of GreedyGroupPOTC, we have that*

$$|V_t^{g,i}| \leq |\mathcal{G}|$$

and hence that $\max_{g \in \mathcal{G}} |V_t^{g,i}| \leq |\mathcal{G}|$.

Proof. Fix an arbitrary bucket $i \in [m]$. We proceed via contradiction: at time $T = t_0$, assume there exists a group g_1 with absolute bias $|\mathcal{G}| + \epsilon$ for some $\epsilon > 0$. Consider $t_1 = (\min\{t \in [T] : \max_{g \in \mathcal{G}} |V_t^{g,i}| = |\mathcal{G}| + \epsilon\}) - 1$, or the iteration preceding the first time step at which the max absolute bias reached $|\mathcal{G}| + \epsilon$. At this iteration, group g_1 must have absolute bias at least $(|\mathcal{G}| - 1) + \epsilon$. For g_1 to have increased its bias after iteration t_1 , there must exist some other group g_2 that also has absolute bias at least that of g_1 , though with opposite sign. For simplicity, we will assume both g_1, g_2 have $|V_{t_1}^{g_1,i}| = |V_{t_1}^{g_2,i}| = (|\mathcal{G}| - 1) + \epsilon$, and that

only two groups have bias $(|\mathcal{G}| - 1) + \epsilon$; more than two groups having absolute bias strictly greater than $|\mathcal{G}| - 1$ will only help us.

Now, consider the iterations before t_1 . Note both g_1, g_2 can't have reached $(|\mathcal{G}| - 1) + \epsilon$ bias for the final time on the same iteration, as they have biases of opposite signs. WLOG, say g_2 reached $(|\mathcal{G}| - 1) + \epsilon$ for the final time after g_1 . Let t_2 be the iteration preceding the final time g_2 reached bias $(|\mathcal{G}| - 1) + \epsilon$; at this time, g_2 must have absolute bias $|V_{t_2}^{g_2, i}| \in [(|\mathcal{G}| - 2) + \epsilon, (|\mathcal{G}| - 1) + \epsilon)$. In order for its bias to have increased, there must exist a different group g_3 which has bias at least that of g_2 (note $g_1 \neq g_3$ as t_2 is after the final iteration at which g_1 reached $(|\mathcal{G}| - 1) + \epsilon$ bias). By assumption, g_3 has bias at most that of g_1 , and again for simplicity we will assume the worst-case scenario when g_2, g_3 both have bias $|V_{t_2}^{g_2, i}| = |V_{t_2}^{g_3, i}| = (|\mathcal{G}| - 2) + \epsilon$.

We recursively continue this procedure for each $3 \leq j \leq |\mathcal{G}|$: WLOG, we assume g_{j-1} reached bias $(|\mathcal{G}| - j + 1) + \epsilon$ for the final time before g_j . Then, denote $t_j > t_{j-1}$, $t_j < t_1$ to be the iteration preceding the final time g_j reached bias $(|\mathcal{G}| - j + 1) + \epsilon$, at which point group g_j and some other group g_{j+1} must have both had bias at least $(|\mathcal{G}| - j) + \epsilon$.

Ultimately, at time $t_{|\mathcal{G}|}$, there must exist distinct groups with biases $(|\mathcal{G}| - j) + \epsilon$ for each $j \in [|\mathcal{G}|]$, with at least two groups with bias in the interval $[\epsilon, 1 + \epsilon)$. But this would require at least $|\mathcal{G}| + 1$ distinct groups, which is a contradiction. $\square$

[GreedyGroupPOTC guarantee] For $m = \sqrt{T/|\mathcal{G}|}$, GreedyGroupPOTC outputs predictions $p = p^{1:T}$ that, against any sequence of adversarially-generated contexts and labels $y = y^{1:T}$, fulfills

$$\text{MCDist}(p, y) \leq 2\sqrt{T|\mathcal{G}|}$$

Proof. Applying Lemma 3.6 to Observation 3,

$$\text{MCDist}(\tilde{p}, y) \leq \sum_{i \in [m]} \max_{g \in \mathcal{G}} |V_T^{g, i}| \leq m|\mathcal{G}|$$

Following the approach from Theorem 3.5, we use the triangle inequality to obtain

$$\text{MCDist}(p, y) \leq \|p - \tilde{p}\|_1 + \text{MCDist}(\tilde{p}, y) \leq \frac{T}{m} + m|\mathcal{G}| \leq 2\sqrt{T|\mathcal{G}|}$$

where the final bound is obtained by setting $m = \sqrt{\frac{T}{|\mathcal{G}|}}$. $\square$

4 Conclusion & Future Work

The algorithms presented in this paper pose a clear tradeoff between the size of the collection of groups and rate of decay with respect to T : Theorem 3.5 guarantees MCDist proportional to $\ln |\mathcal{G}|$ decaying like $T^{3/4-1}$, while Theorem 3.4 obtains MCDist decaying faster at $T^{1/2-1}$ but proportional to $\sqrt{|\mathcal{G}|}$. It remains to find an algorithm obtaining the best of both worlds, which would fulfill an original goal of an algorithm obtaining MCDist bounded by $O(\sqrt{T \ln |\mathcal{G}|})$.

With the $O(\sqrt{T \ln |\mathcal{G}|})$ bound in mind, some possible extensions of this project are:

- **Extending POTC-Cal to POTC-Multical:** The algorithm described in [1] utilizes the power of two choices algorithm in [5]. In particular, the option of having two choices that differ only by a small sliver pairs nicely with the ℓ_1 metric we are optimizing for. An extension of this algorithm to the multicalibration setting would be of great help in bounding the error of the look ahead predictions.
- **ECE on All Subsequences:** Fundamentally, a key difference between the aforementioned multicalibration algorithm and our intent here is the objective function: multicalibration as defined by [8] is a guarantee on all groups and all buckets, so the multicalibration algorithm minimizes $\max_{g \in \mathcal{G}} \max_{i \in [m]} |V_s^{g,i}|$. On the other hand, we seek to minimize the ECE conditioned on each group, i.e. $\max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}|$. Although the multicalibration algorithm obtains $O(\sqrt{T \ln(|\mathcal{G}|m)})$ multicalibration error, a factor of m is lost to convert this guarantee into low ECE for each group, thereby giving us our $O(T^{3/4})$ bound.
We note the UnbiasedPrediction algorithm from [6] which uses a modified multiplicative weights algorithm from [3] guarantees bias $\alpha(E) = O(\frac{\ln(T|\mathcal{E}|)}{\sqrt{T_E}})$ for a collection of events $\mathcal{E}$. For our purposes, $\mathcal{E} = \{E := E(g, i); g \in \mathcal{G}, i \in [m]\}$.
- **LowerMulticalibrationDistance:** The original analysis in [7] proved the existence of a $O(\sqrt{T})$ calibration distance algorithm using the notion of a LowerCalibrationDistance to tightly bound calibration distance. It is very likely a similar argument can be used to show the existence of a low-error multicalibration distance algorithm.
- **Lower bound:** Again with reference to [7], it remains to show the lower bound of $O(T^{1/3} \ln |\mathcal{G}|)$ applies to multicalibration distance.

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